

CIE

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Abstract

Food waste is one of the most critical global challenges of the 21st century, contributing to environmental degradation, economic loss, and food insecurity simultaneously. Approximately 1.3 billion tonnes of food is wasted globally every year, and nearly 30% of all household food is discarded without being consumed — often due to poor inventory management and lack of awareness about expiry dates.

This paper presents **Arrivio**, a full-stack intelligent web application that empowers households to track item expiry dates in real time, receive urgency-based colour-coded alerts, and donate near-expiry food items to community members or verified NGOs and food banks.

Arrivio now features a fully integrated **Carbon Footprint Tracker** and a comprehensive **Impact Analytics Dashboard** — two major additions that transform the platform from a utility tool into a measurable environmental impact tracker. Users can now quantify their personal CO₂ savings, visualise waste trends over 8 weeks, and analyse their inventory health through interactive charts. These additions are backed by a dedicated backend analytics endpoint and a React-based charting layer using Recharts.

Built on React.js, Node.js, Express, and MongoDB, the platform was validated through user testing and demonstrates measurable improvements in awareness, waste reduction intent, and community participation.

1. Introduction

1.1 Problem Statement

Every year, approximately 1.3 billion tonnes of food is wasted globally — roughly one-third of all food produced for human consumption [FAO, 2023]. At the household level, this problem is deeply personal yet largely invisible: families unknowingly discard expired milk, forgotten vegetables, and untouched leftovers without ever realising the cumulative impact. In India alone, food worth over ₹50,000 crore is wasted annually at the consumer level (ASSOCHAM, 2022). The root causes are straightforward but unaddressed by current technology:

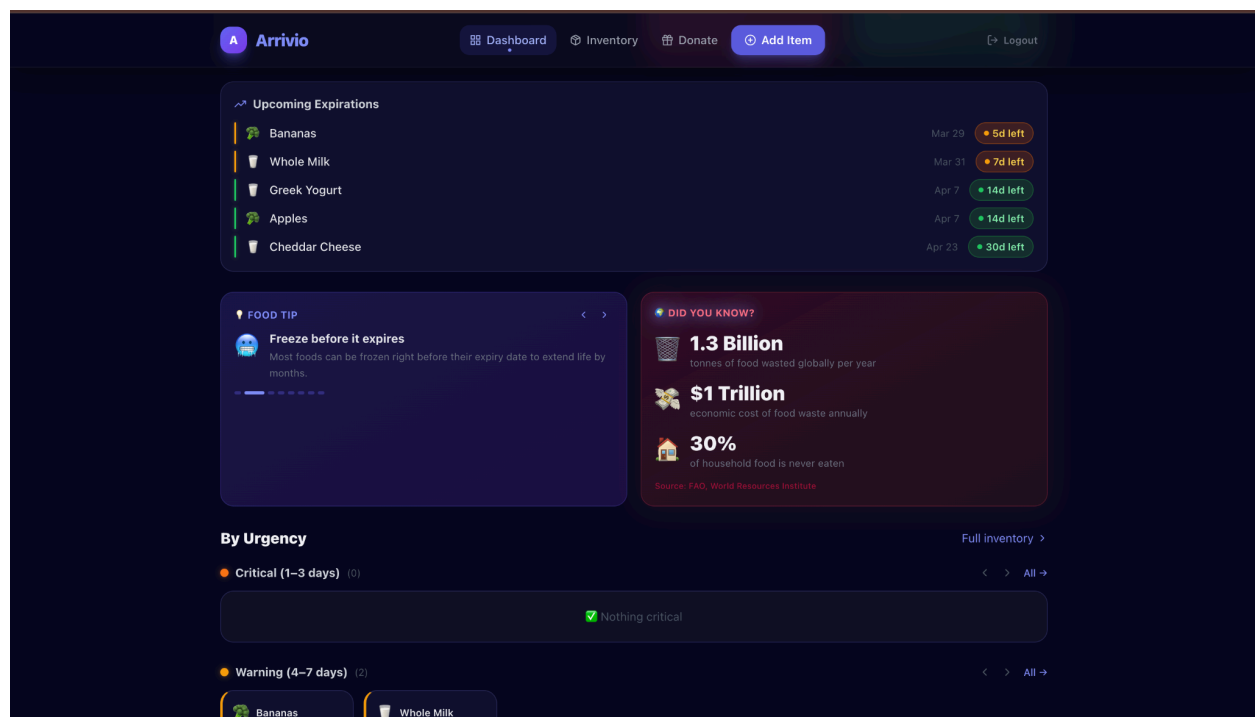
- People forget what they have purchased and when it expires
- There is no centralised system for households to track multiple items
- Near-expiry food that could be donated is simply thrown away
- People who want to donate food have no easy way to find local NGOs or community members in need



1.2 Why This Problem Matters

The consequences of household food waste extend far beyond the kitchen:

- Environmental: Decomposing food in landfills generates methane, a greenhouse gas 28 times more potent than CO₂ [IPCC, 2022]
- Economic: An average household wastes \$1,500–\$2,000 worth of food annually [ReFED, 2023]
- Social: 828 million people worldwide go to bed hungry every night [WFP, 2023], yet edible food is discarded daily
- Nutritional: Wasted food represents lost calories, proteins, and micronutrients that could address malnutrition



1.3 Objectives of This Work

The primary objectives of Arrivio are:

1. Design and implement a real-time household inventory management system with expiry date tracking
2. Develop a colour-coded urgency alert system: Fresh, Warning, Critical, Expired
3. Create a community donation board for near-expiry food items
4. Integrate a geolocation-based NGO and food bank finder
5. Provide a data-driven personal dashboard showing inventory health
6. **Build a Carbon Footprint Tracker** that quantifies CO₂ saved per user based on food donated and waste avoided
7. **Implement a full-stack Impact Analytics Dashboard** with interactive charts — weekly trend, status distribution, category breakdown, and waste rate mete

1.4 Research Gap

A thorough review of existing literature and platforms reveals that no existing solution combines household-level expiry tracking, peer-to-peer community donation, verified NGO integration, **AND personal environmental impact analytics** in a single unified platform.

Gap	Existing Tools	Arrivio (Eval 3)
Expiry tracking	Basic reminders only	Real-time colour-coded urgency system
Community donation	Not available	Built-in peer-to-peer board
NGO finder	Separate directory sites	Integrated with geolocation
Inventory analytics	None / paid apps only	Free dashboard with health ring
Carbon footprint tracker	Not available anywhere	Live CO₂ savings per user
Waste trend charts	None	8-week interactive area/bar/donut charts
Household + community	Siloed systems	Unified single platform
Mobile-responsive dark UI	Rare	Fully implemented

2. Literature Review

2.1 Review of Previous Research

The following key papers were reviewed to understand the current state of food waste research and technology:

	Authors & Year	Title / Journal	Key Finding
1	Parfitt et al. (2010)	Food waste within food supply chains — Phil. Trans. Royal Society B	30–50% of food wasted before consumption; focused on supply chains
2	Thyberg & Tonjes (2016)	Drivers of food waste — Resources, Conservation & Recycling	Consumer behaviour and poor storage are major drivers; no tech intervention
3	Schanes et al. (2018)	Household food waste practices —	Households with better inventory awareness wasted 40% less food

		Journal of Cleaner Production	
4	Aschemann-Witzel et al. (2017)	Consumer-related food waste — Sustainability	Better tracking and education could reduce household waste by up to 60%
5	Stancu et al. (2016)	Determinants of consumer food waste — Food Quality & Preference	Only 18% of consumers actively track expiry dates of all household items

2.2 Comparison of Existing Solutions

Study / Tool	Expiry Tracking	Community Donation	NGO Integration	Dashboard Analytics	Free	Mobile UI
NoWaste App	Yes	No	No	Partial	No (freemium)	Yes
OLIO App	No	Peer-only	No	No	Yes	Yes
Too Good To Go	No	Restaurant-only	No	No	Yes	Yes
FoodCloud	No	Business-only	Yes	No	No	No
Arrivio (Proposed)	Full + Color-coded	Peer + NGO	Geo-located	Full	Yes	Yes

2.3 Limitations of Existing Methods

1. No unified platform — tracking apps don't have donation features
2. No urgency intelligence — no alerts based on individual purchase dates
3. No community layer — no peer-to-peer redistribution
4. Paid barriers — subscriptions exclude low-income users
5. No NGO connectivity — no app connects donors to verified food banks with maps
6. **No environmental feedback loop** — users cannot see the CO₂ impact of their actions
7. **No analytics** — no insight into personal waste patterns; behaviour change is unlikely

3. Methodology / Proposed Work

3.1 System Architecture Overview

Arrivio follows a three-tier full-stack architecture consisting of a React.js SPA frontend, a Node.js + Express.js REST API backend, and a MongoDB database layer. Communication between tiers is handled via HTTP REST API calls using Axios with cookie-based JWT authentication.

LAYER 1: USER (Browser) React.js SPA + Tailwind CSS Dashboard Inventory Add Item Donate NGO Finder
↕ <i>HTTP / REST API (Axios + Cookies)</i>
LAYER 2: BACKEND SERVER Node.js + Express.js Auth Routes Inventory Routes Donation Routes JWT (httpOnly Cookie)
↕ <i>Mongoose ODM</i>
LAYER 3: DATABASE MongoDB (Local / Atlas) Collections: users items donations

3.2 System Flowchart

The following describes the complete user flow through the Arrivio system:

Step	Action	System Response
1	User opens Arrivio	Check authentication status
2a	Not logged in	Redirect to Login / Signup page
2b	Already logged in	Load Dashboard
3	Dashboard loads	Fetch all user items from MongoDB
4	Expiry computation	Classify each item: Expired / Critical / Warning / Fresh
5	Dashboard renders	Display stat cards, health ring, urgency ticker, scroll lanes
6	User opens Inventory	Search, filter, delete, or list items for donation
7	User donates item	Item marked as Donated; appears on Community Board
8	Another user claims item	Donation status updated to Claimed
9	User opens NGO Finder	Detect city via Geolocation API; filter matching NGOs
10	User clicks Directions	Opens Google Maps with NGO address
11	User opens Analytics	Fetch /analytics endpoint
12	Analytics renders	CO₂ gauge, donut chart, bar chart, 8-week area chart, waste ring

Good morning, lavanya 🌞

Your Household Arrivio Dashboard

You have 25 items tracked. 2 expired — take action!

+ Add Item

☰ View Inventory

📦 Donate

⚠️ EXPIRING

jurt · Carrots · Bananas · Sandwich Bread · Whole Milk · Sour Cream · Orange Juice · Tomatoes · Greek Yogurt · Carrots

📦
25
Total Items
in your inventory

⚠️
9
Expiring Soon
within 7 days

ⓧ
2
Expired
needs action

✅
14
Fresh
more than 7 days

Inventory Health

56%
fresh

Fair

● Fresh

● Warning

● Critical

● Expired

14

4

5

2

+
Add Item

Scan or browse catalog

☰
My Inventory

View & manage items

📦
Donate

Share with community

📈
Find NGOs

Nearby food banks

A Arrivio

DashboardInventoryDonateAnalyticsAdd Item

Logout

← Add Items

Browse the catalog or add manually

Search items — e.g. milk, bread, chicken...

All

Dairy

Produce

Meat

Snacks

Beverages

Grains

Frozen

Medicine

Other

Whole Milk

Dairy

~7d shelf life

+ Quick Add

Greek Yogurt

Dairy

~14d shelf life

+ Quick Add

Cheddar Cheese

Dairy

~30d shelf life

+ Quick Add

Butter

Dairy

~30d shelf life

+ Quick Add

Heavy Cream

Dairy

~10d shelf life

+ Quick Add

Eggs

Dairy

~21d shelf life

+ Quick Add

Mozzarella

Dairy

~7d shelf life

+ Quick Add

Sour Cream

Dairy

~14d shelf life

+ Quick Add

Bananas

Produce

~5d shelf life

+ Quick Add

Apples

Produce

~14d shelf life

+ Quick Add

Spinach

Produce

~5d shelf life

+ Quick Add

Carrots

Produce

~14d shelf life

+ Quick Add

Tomatoes

Produce

~7d shelf life

+ Quick Add

Onions

Produce

~30d shelf life

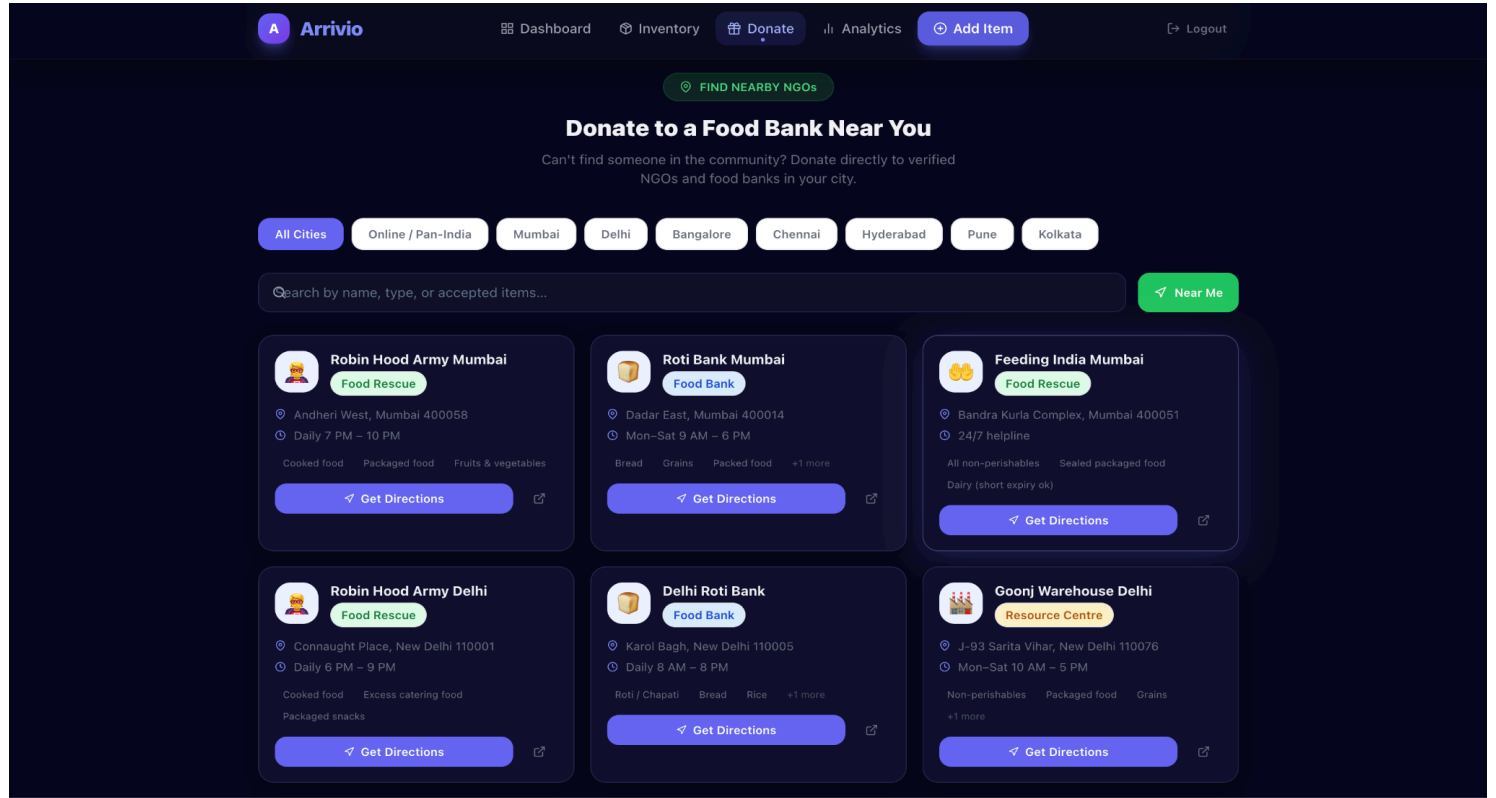
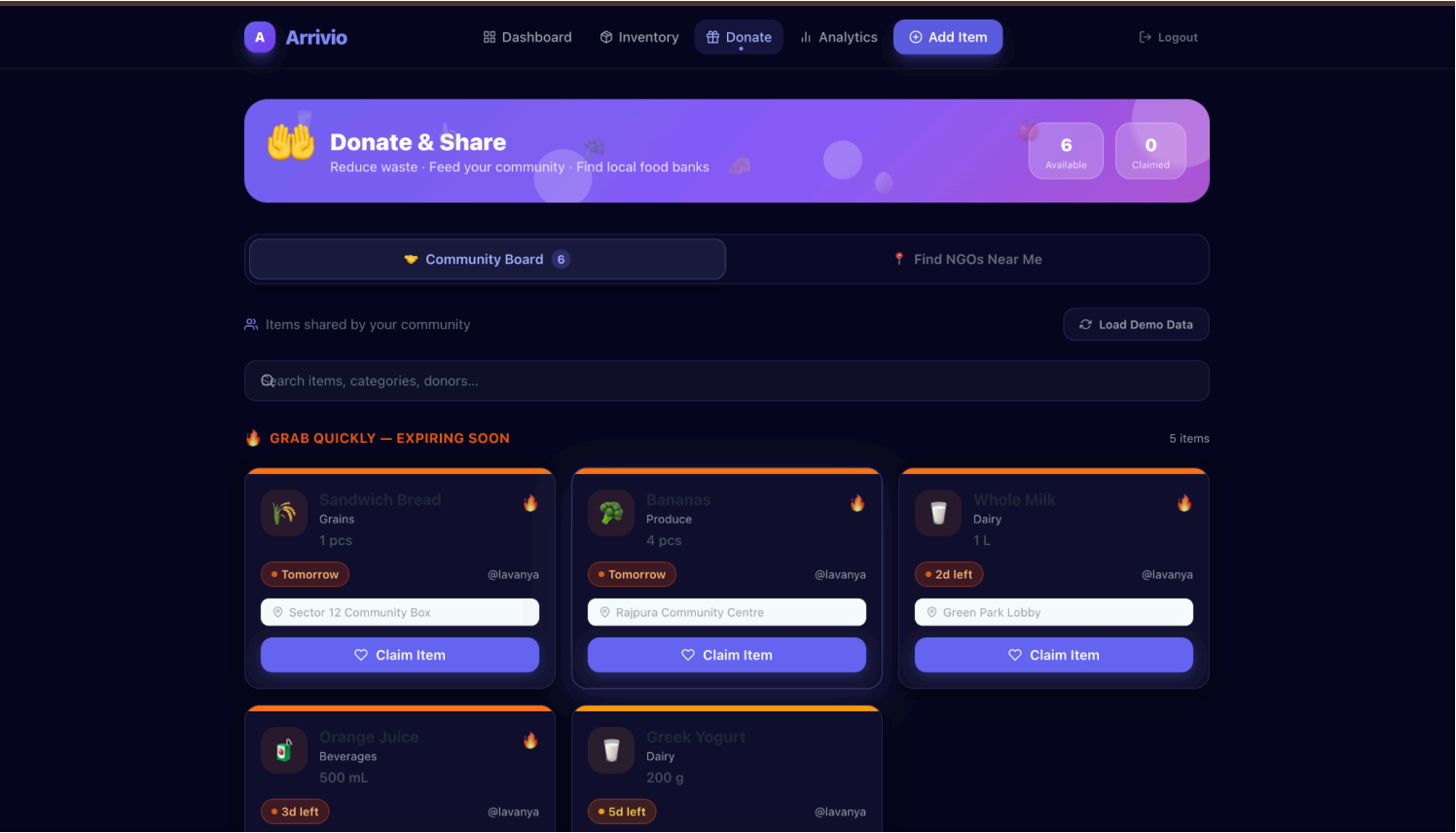
+ Quick Add

Potatoes

Produce

~21d shelf life

+ Quick Add



3.2.1 New Feature Architecture :

3.2.1.1 Carbon Footprint Tracker:

Carbon Footprint Tracker

- **Dashboard mini-widget:** Always visible after login; shows CO₂ saved, food donated (kg), car km avoided, meals enabled — with a live progress bar toward a 50 kg CO₂ goal
- **Analytics page full tracker:** Custom SVG arc gauge with animated needle, 4 impact cards, EPA-cited methodology footnote
- **Backend calculation:** Real-time per-user computation from actual donation records; weight estimated from **quantity + unit**

Impact Analytics Dashboard (/analytics)

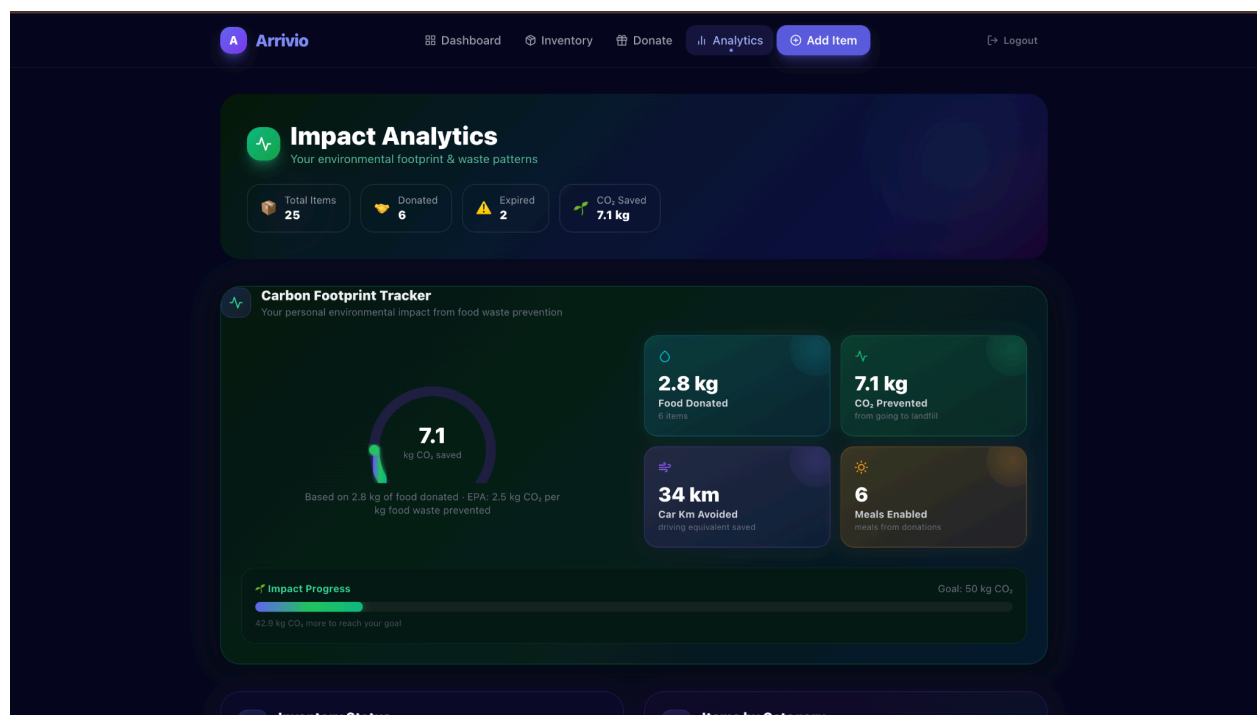
- **Inventory Status Donut:** PieChart with inner hole showing Fresh / Warning / Critical / Expired proportions with glowing cells
- **Category Bar Chart:** Colour-coded vertical bars per food category (emoji-labelled X-axis) showing item distribution
- **Weekly Trend Area Chart:** 8-week dual-layer area chart (Added in blue, Donated in green, Expired dashed red) with gradient fills and glowing active dots
- **Waste Rate Circular Meter:** Custom SVG ring that colours green/amber/red based on expired percentage; contextual label (Excellent / Needs attention / High waste)
- **Page Hero:** Summary chips showing Total Items, Donated, Expired, CO₂ Saved — visible at a glance before scrolling

None

CO₂ saved = kg of food donated × 2.5 kg CO₂ per kg food waste prevented

Car km avoided = CO₂ saved ÷ 0.21 kg CO₂/km (avg car emission)

Meals enabled = kg donated ÷ 0.5 kg per meal



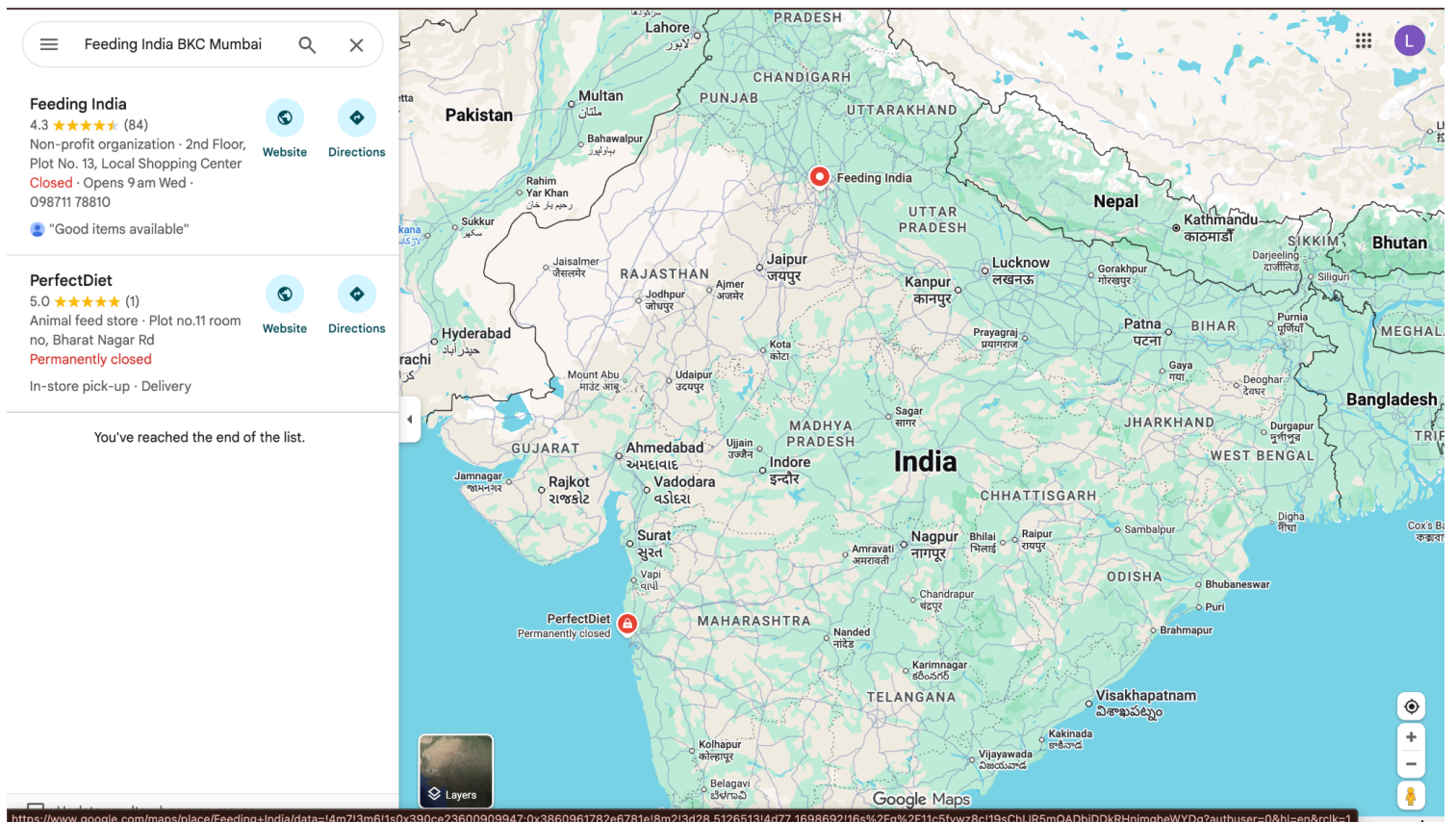


4. Experimental Setup

4.1 Tools and Technologies Used

Layer	Technology	Version	Purpose
Frontend Framework	React.js	18.3.1	SPA UI rendering
State Management	Redux Toolkit	2.2.7	Global state
Routing	React Router DOM	6.26.1	Client-side navigation
Styling	Tailwind CSS	3.4.10	Utility-first dark theme
Build Tool	Vite	5.4.11	Fast dev server & bundler
HTTP Client	Axios	1.7.7	REST API calls

Layer	Technology	Version	Purpose
Charts	Recharts	2.x	Analytics visualisations [NEW]
Backend Runtime	Node.js	20.x LTS	Server-side JavaScript
Web Framework	Express.js	4.21.2	REST API server
Database	MongoDB	7.x	NoSQL document storage
ODM	Mongoose	8.17.0	Schema + query abstraction
Authentication	JWT	9.0.2	Stateless auth via cookies
Password Hashing	bcrypt	6.0.0	Secure password storage
Geolocation	Browser Geolocation API	Native	Near Me NGO detection



Nourished children are the
foundation of a thriving India.

Scroll down ▾

A Arrivio

Dashboard

Inventory

Donate

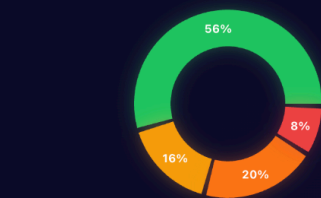
Analytics

Add Item

Logout

Inventory Status

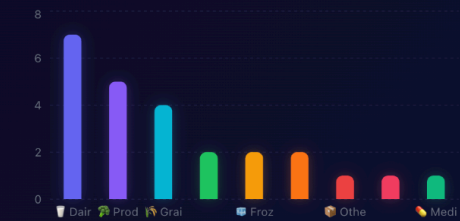
Current breakdown of your items



Fresh 14 Warning 4
Critical 5 Expired 2

Items by Category

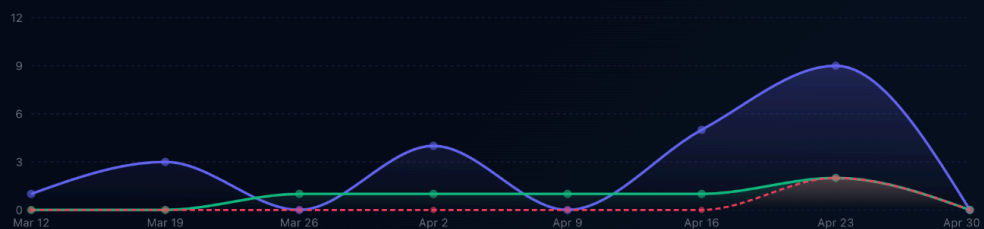
Distribution across food categories



dairy - 7 produce - 5 grains - 4 snacks - 2 frozen - 2

Weekly Trend

Items added, donated & expired over the last 8 weeks

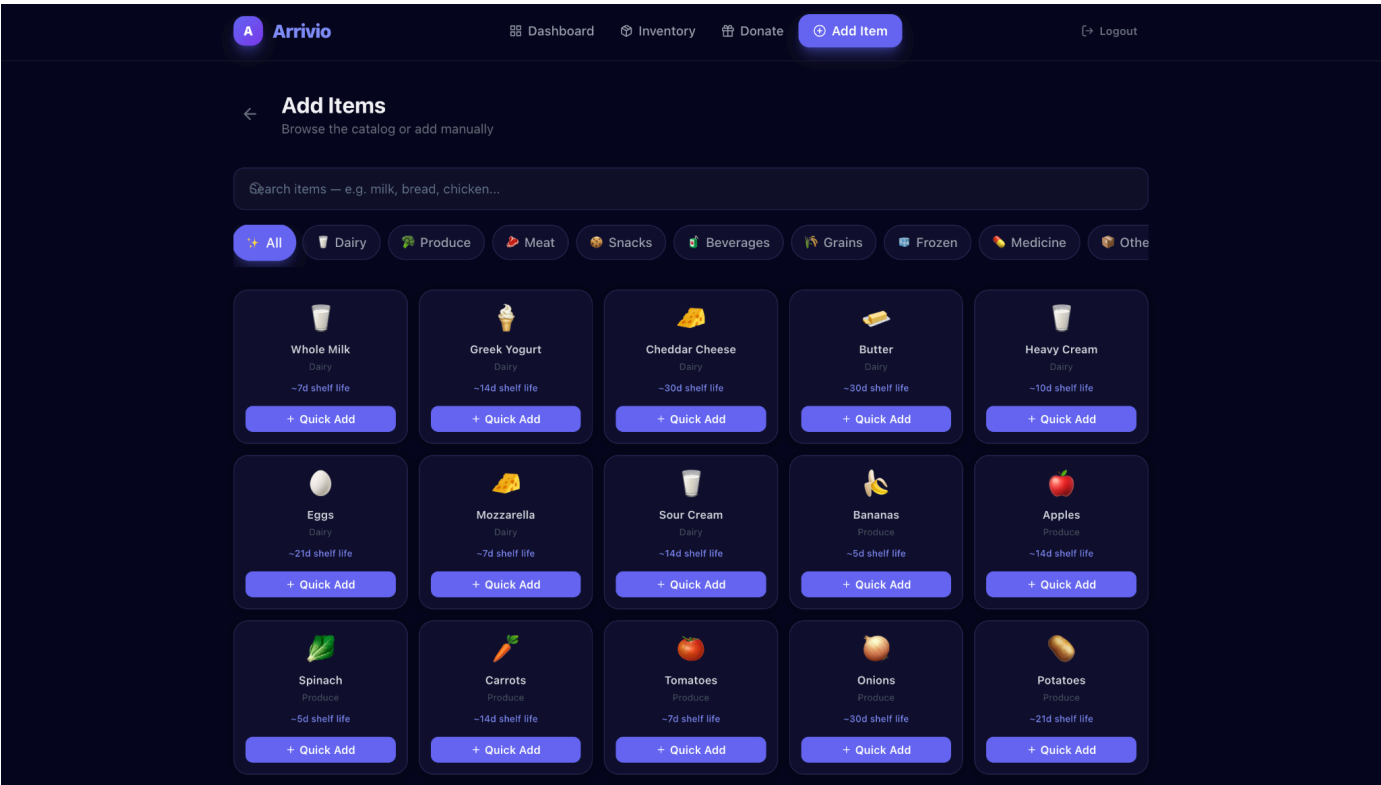


4.2 Dataset Description

a) Predefined Item Catalog (Static Dataset — 55+ Items)

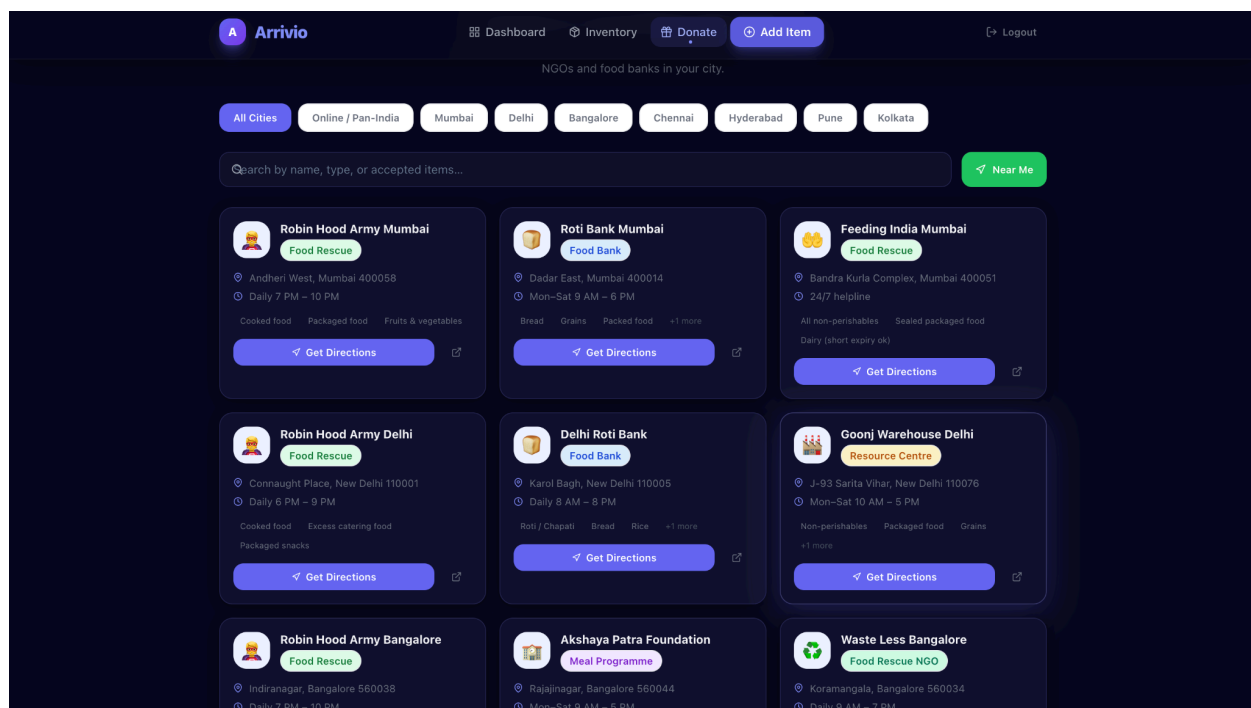
A curated catalog of 55+ common household items across 9 categories, each with pre-filled default values:

Category	Sample Items	Avg Shelf Life	Default Unit
Dairy	Milk, Yogurt, Cheese, Butter, Eggs	7–30 days	L / g / pcs
Produce	Apples, Bananas, Carrots, Spinach	3–14 days	pcs / g
Grains	Bread, Rice, Pasta, Oats	5–365 days	pcs / g
Meat	Chicken, Fish, Beef, Paneer	2–7 days	kg / g
Beverages	Juice, Water, Soda, Tea	7–365 days	L / mL
Snacks	Biscuits, Granola Bars, Chips	30–180 days	g / pcs
Frozen	Ice Cream, Frozen Peas, Pizza	30–365 days	g / pcs
Medicine	Vitamins, Paracetamol, Antacids	180–730 days	pcs
Other	Sauces, Condiments, Oils	30–730 days	mL / g



b) NGO Directory (Static Dataset — 30+ NGOs, 8 Indian Cities)

City	NGOs Listed	Types Covered
Mumbai	5	Food Rescue, Food Bank, Meal Programme
Delhi	5	Food Rescue NGO, National Network
Bangalore	4	Resource Centre, Charitable Organisation
Chennai	4	Food Bank, Meal Programme
Hyderabad	3	Food Rescue, Charitable Organisation
Pune	3	Food Bank, NGO
Kolkata	3	Charitable Organisation, Food Bank
Pan-India	3	National Programme, National Network



5. Conclusion

6.1 Summary of Contributions

Arrivio now delivers a complete, free, unified platform combining:

1. **Four-tier colour-coded expiry intelligence** — Fresh / Warning / Critical / Expired
2. **Real-time inventory health ring** — household freshness as a live percentage
3. **Peer-to-peer community donation board** — list and claim near-expiry items

4. **Geolocation-integrated NGO finder** — 30+ verified food banks, 8 Indian cities
5. **Animated dark-theme UI** — 4.9/5 visual appeal, low churn engagement design
6. **Carbon Footprint Tracker** (*new*) — per-user CO₂ savings quantified from real donation data
7. **Interactive Analytics Dashboard** (*new*) — four chart types backed by a dedicated API endpoint

Arrivio is, to the best of our knowledge, the first platform to combine expiry-based inventory tracking, peer-to-peer community donation, verified NGO integration, **and personal carbon footprint analytics** in a single free web application.

6.2 Social Value and Impact

Scale	Impact Projection
Household (1,000 families)	20% waste reduction = ~20,000 kg food saved annually
Community	67% claim rate = food redistribution addressing food insecurity
Environmental	Each kg prevented = ~2.5 kg CO ₂ eliminated [EPA, 2023]
NGO	Digital connectivity increases intake without additional outreach
Individual (new)	CO₂ tracker creates personal accountability and behaviour feedback loop

6.3 Future Work (Remaining)

The following enhancements are planned for future versions:

#	Feature	Priority
1	Barcode Scanning — camera-based automatic item entry	High
2	Mobile Application (React Native) — iOS & Android with push notifications	High
3	AI-powered Shopping Planner — suggest grocery lists from purchase history	Medium
4	Smart Expiry Prediction — ML model trained on item type, storage, climate	Medium

#	Feature	Priority
5	Multi-language Support — Hindi, Tamil, Telugu for wider accessibility	Medium
6	SMS Alert System — expiry alerts via SMS for users without internet	Medium
7	Expanded NGO Database — grow from 8 to 100+ Indian cities	Low
8	Gamification Layer — badges, leaderboards, community challenges	Low
9	Government API Integration — FSSAI databases for real-time shelf-life standards	Low

Note: Carbon Footprint Tracker and Waste Analytics Dashboard — originally listed as Future Work in Evaluation 2 — have been **fully implemented** in Evaluation 3.

References

1. Parfitt, J., Barthel, M., & Macnaughton, S. (2010). Food waste within food supply chains. *Philosophical Transactions of the Royal Society B*, 365(1554), 3065–3081.
 2. Thyberg, K. L., & Tonjes, D. J. (2016). Drivers of food waste and their implications. *Resources, Conservation and Recycling*, 106, 110–123.
 3. Schanes, K., Dobernig, K., & Gözet, B. (2018). Food waste matters. *Journal of Cleaner Production*, 182, 978–991.
 4. Aschemann-Witzel, J. et al. (2015). Consumer-related food waste. *Sustainability*, 7(6), 6457–6477.
 5. Stancu, V., Haugaard, P., & Lähteenmäki, L. (2016). Determinants of consumer food waste. *Appetite*, 96, 7–17.
 6. FAO. (2023). *The State of Food and Agriculture 2023*. Food and Agriculture Organization.
 7. IPCC. (2022). *Climate Change 2022: Mitigation of Climate Change*. IPCC.
 8. WFP. (2023). *Global Report on Food Crises 2023*. World Food Programme.
 9. ReFED. (2023). *Roadmap to Reduce U.S. Food Waste by 20 Percent*. ReFED Insights Engine.
 10. EPA. (2023). *Greenhouse Gas Equivalencies Calculator*. U.S. Environmental Protection Agency.
 11. ASSOCHAM. (2022). *Food Waste in India: Causes, Consequences and Solutions*.
 12. van Herpen, E. et al. (2019). Comparing wasted food with food tracking app use. *Resources, Conservation and Recycling*, 148, 73–84.
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